

Digital Experience Measurement Capability

1. Executive Architecture Statement

This proposal defines an enterprise Digital Experience Measurement Capability using UK Open Banking (OBIE / PSD2) as the first implementation pilot and proof point.

The capability objective is not to create an Open Banking reporting solution.

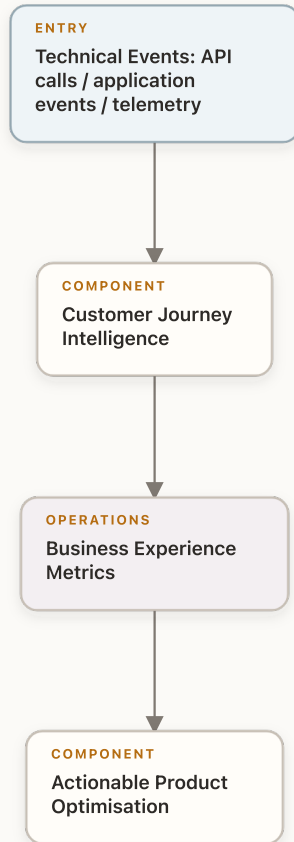
The objective is to establish a reusable enterprise architecture capability that bridges:

- Digital Platform Engineering
- API Observability
- Customer Journey Analytics
- Product Experience Optimisation

The strategic transformation is:

SYSTEM DIAGRAM

Strategic Transformation



Strategic Transformation

The target state enables HSBC to understand not only whether systems operate correctly, but whether customers successfully achieve their intended outcomes.

2. Executive Problem Statement

Digital platforms generate extensive operational data:

- API gateway events
- Application telemetry
- Mobile/web interaction events
- Transaction processing events
- Customer behavioural signals

However, these signals are fragmented across technology domains.

Current challenges:

Lack of end-to-end journey visibility

Example:

A payment API is healthy, but customers abandon the payment journey because authentication takes too long.

Traditional monitoring sees:

```
API availability: 99.9%  
Latency: acceptable
```

Customer experience sees:

```
Payment completion reduced by 12%  
Customer abandonment increased
```

Lack of consistent measurement model

Different platforms define:

- different events
- different tags
- different KPIs
- different reporting approaches

This prevents enterprise comparison.

Limited insight-to-action capability

Existing reporting answers:

"What happened?"

The target capability answers:

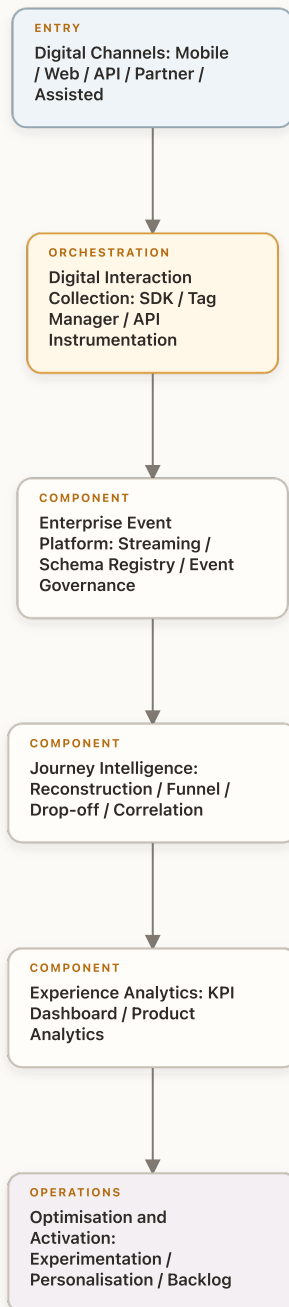
"Why did it happen and what should we change?"

3. Architecture Vision

Digital Experience Intelligence Platform

SYSTEM DIAGRAM

Digital Experience Intelligence Platform



Digital Experience Intelligence Platform

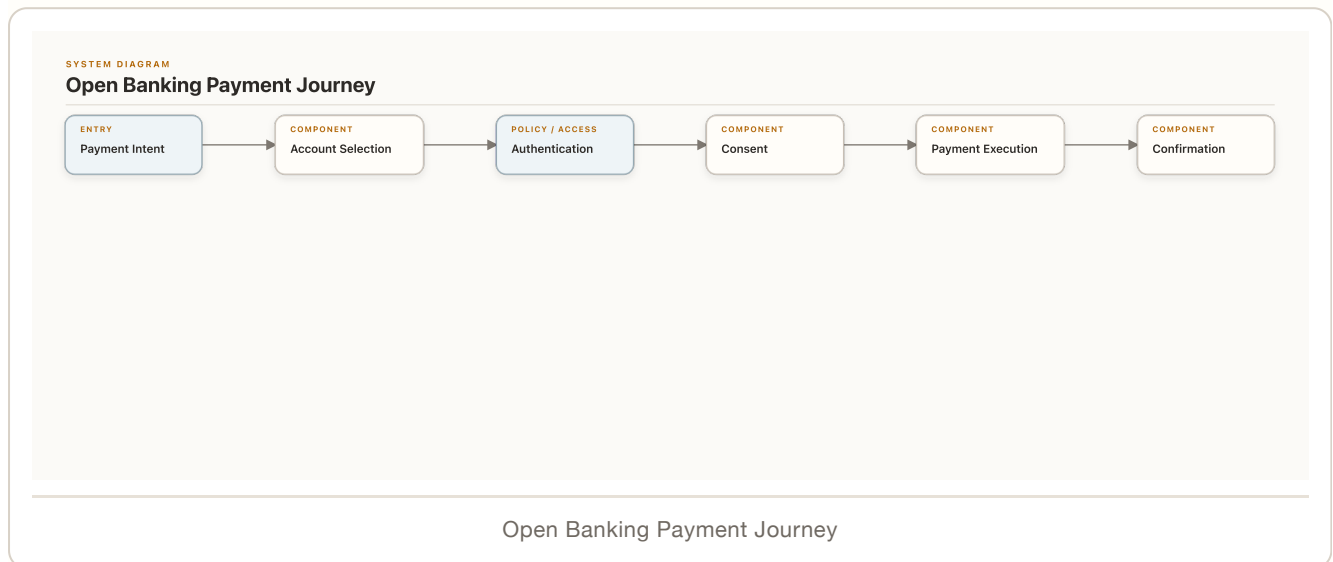
4. Architecture Principles

Principle 1 --- Journey Centric Measurement

The primary unit of measurement is customer journey, not API.

Example:

Open Banking Payment Journey:



Principle 2 --- API Events Must Carry Business Context

A technical event alone is insufficient.

Traditional:

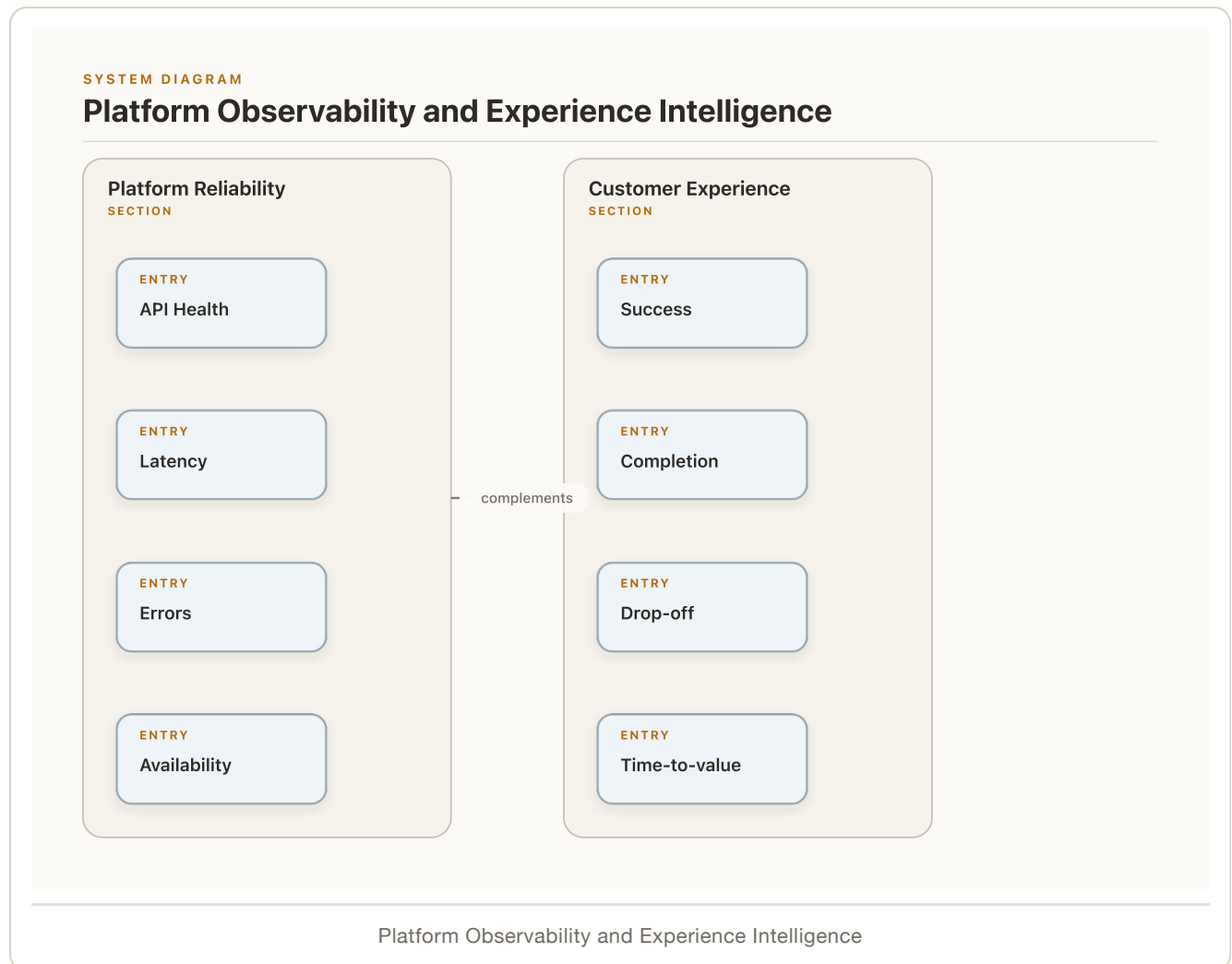
```
{
  "api": "payment-service",
  "status": 500
}
```

Target:

```
{
  "eventName": "payment_failed",
  "journey": "open_banking_payment",
  "stage": "authentication",
  "channel": "mobile",
  "failureReason": "ASPSp_TIMEOUT",
  "customerImpact": "journey_abandonment"
}
```

Principle 3 --- Separate Platform Observability From Experience Intelligence

Two complementary layers:



5. Open Banking Pilot Reference Architecture

UK OBIE Payment Journey

Actors:

- Customer
- HSBC Channel
- HSBC Open Banking Platform
- TPP
- ASPSP
- Analytics Platform

Journey:

SYSTEM DIAGRAM

Open Banking Pilot Reference Journey



Open Banking Pilot Reference Journey

6. API Journey Measurement Model

Event Sources

Channel Events

Examples:

```
PAYMENT_STARTED  
  
CONSENT_PAGE_VIEWED  
  
CONSENT_APPROVED
```

Sources:

- Mobile SDK
- Web Tagging
- Digital Channels

API Events

Examples:

```
POST /consents  
  
POST /payments  
  
GET /accounts
```

Converted into:

```
CONSENT_CREATED  
  
PAYMENT_SUBMITTED  
  
PAYMENT_COMPLETED
```

Operational Events

Examples:

ASPSP_TIMEOUT

HTTP_500

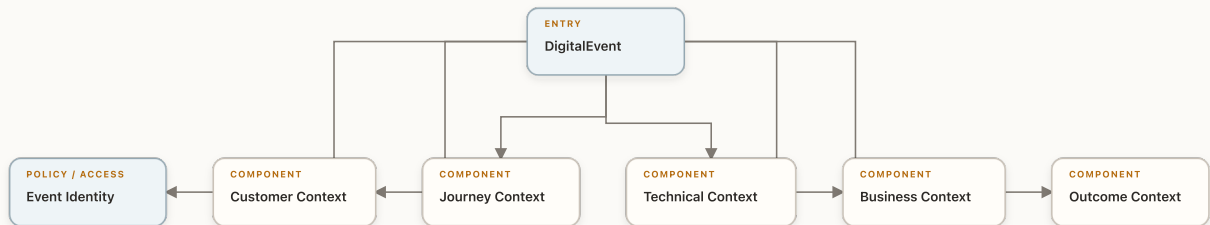
LATENCY_THRESHOLD_BREACH

7. Canonical Enterprise Event Model

All events should follow a common schema.

SYSTEM DIAGRAM

Canonical Enterprise Event Model



Canonical Enterprise Event Model

Example:

```

{
  "eventId": "12345",

  "journey": {
    "name": "OPEN_BANKING_PAYMENT",
    "stage": "AUTHENTICATION"
  },

  "customer": {
    "segment": "RETAIL"
  },

  "technical": {
    "service": "consent-api",
    "latency": 2500
  },

  "outcome": {
    "status": "FAILED",
    "reason": "TIMEOUT"
  }
}

```



8. Journey Intelligence Layer

8.1 Journey Reconstruction

Correlate events into customer journey instances.

Example:

Journey ID: ABC123

10:00 Payment Started

10:01 Bank Selected

10:02 Authentication Started

10:05 Timeout

8.2 Funnel Analytics

Example:

Stage	Volume
Payment Started	100,000
Bank Selected	96,000
Authentication Completed	90,000
Payment Completed	87,000

8.3 API-to-Customer Impact Correlation

The key architecture capability.

Example:

Technical:

ASPSP connector latency increased

Business impact:

Authentication success reduced

Payment completion reduced

Customer abandonment increased

9. Experience Metrics Framework

Executive Metrics

Customer Task Success Rate

Measures whether customers achieve intended outcomes.

Successful journeys

Started journeys

Journey Completion Rate

Measures funnel completion.

Digital Cycle Time

Measures customer time-to-value.

Digital Self-Service Rate

Measures completion without assisted channels.

10. API Event To Business Metric Mapping

API / Event	Journey Stage	Metric	Business Impact
Payment Started	Initiation	Journey Start Volume	Demand
Consent Created	Consent	Consent Completion	Conversion
Authentication Failed	Authentication	Failure Rate	Drop-off
Payment Submitted	Execution	Task Success	Revenue
Payment Confirmed	Completion	Completion Rate	Customer Satisfaction

11. Data Architecture

Bronze Layer

Raw event ingestion.

Sources:

- API Gateway
 - Application Logs
 - Mobile/Web Events
-

Silver Layer

Enterprise canonical events.

Responsibilities:

- Validation
 - Enrichment
 - Correlation
-

Gold Layer

Business intelligence model.

Examples:

```
FACT_CUSTOMER_JOURNEY
```

```
FACT_DIGITAL_EVENT
```

```
FACT_EXPERIENCE_METRIC
```

12. Technology Capability Mapping

Potential implementation components:

Collection

- Tealium
- SDK instrumentation
- API Gateway event capture

Streaming

- Kafka
- Event Hub
- Cloud Pub/Sub

Processing

- Stream processing
- Event correlation engine

Analytics

- Looker
- Product analytics platform

Activation

- Personalisation engines
 - Experimentation platforms
-

13. Governance Operating Model

Journey Owner

Accountable for:

- Journey definition
- Customer outcome
- KPI targets

Digital Platform Owner

Accountable for:

- Event framework
- Tagging standards
- Platform capability

Engineering

Accountable for:

- Instrumentation
- Event quality

Data & Analytics

Accountable for:

- Data models
 - Insights
 - Reporting
-

14. Security and Compliance Considerations

Open Banking requires:

- PSD2 compliance
- Consent management
- Customer data protection
- Data minimisation
- Auditability

Design principles:

- No unnecessary PII in events
 - Tokenised customer identifiers
 - Role-based access
 - Data retention controls
-

15. Delivery Roadmap

Phase 1 --- Foundation

Deliver:

- OB journey model
- Event taxonomy
- API instrumentation
- Canonical schema

Phase 2 --- Intelligence

Deliver:

- Journey analytics
- KPI dashboards
- Drop-off analysis

Phase 3 --- Optimisation

Deliver:

- Root cause analytics
 - Experiment measurement
 - Continuous improvement loop
-

16. Enterprise Scaling Strategy

After Open Banking validation:

Apply the same capability to:

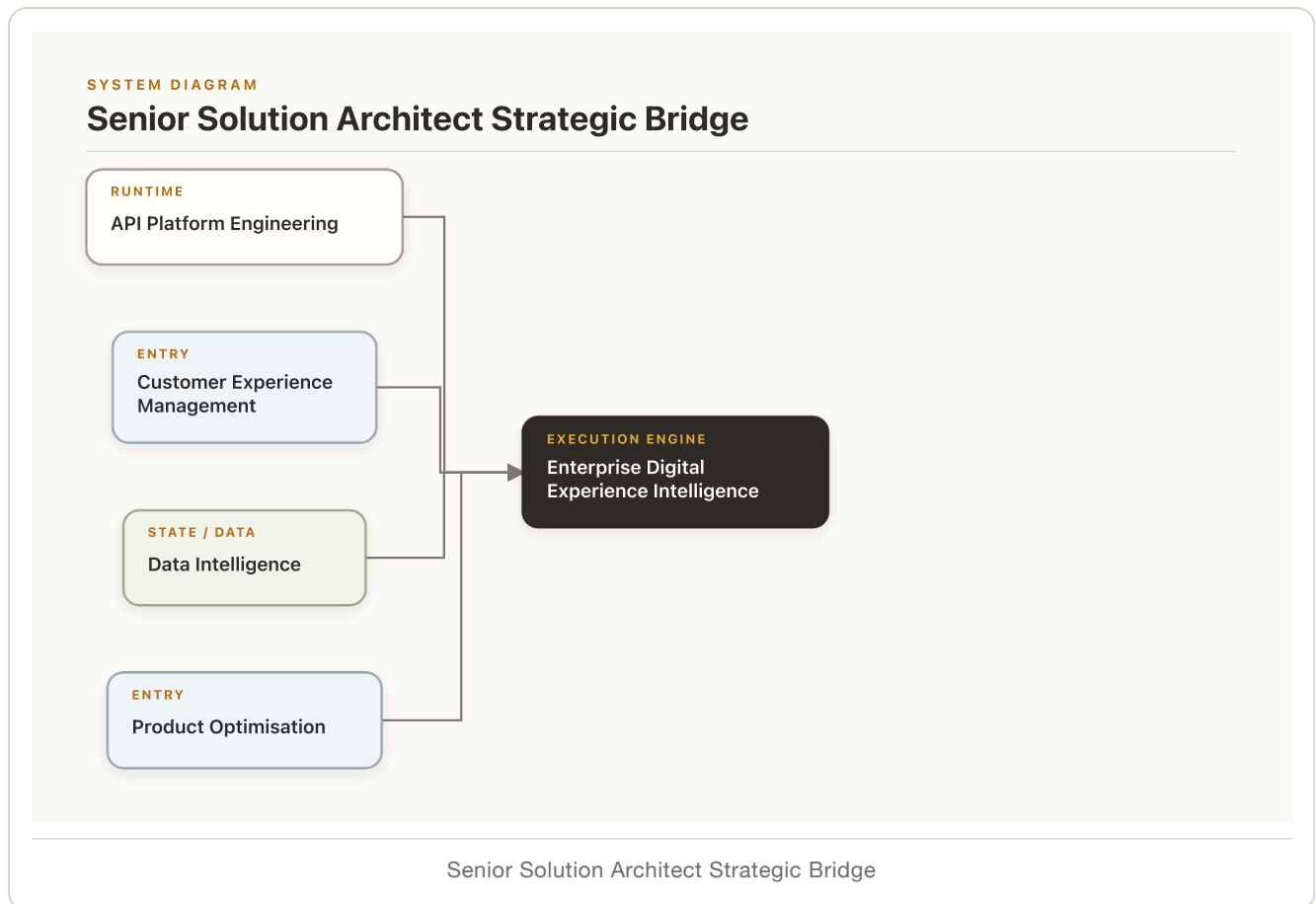
- Retail account opening
- CMB onboarding
- CIB payment approval
- Wealth investment
- Lending journeys

The long-term target:

Enterprise Digital Experience Intelligence Platform

17. Senior Solution Architect Positioning

The architecture role is to create the strategic bridge:



The core proposition:

"Transform API-level interaction events into enterprise customer journey intelligence, enabling measurable digital experience improvement across HSBC platforms."